

Hanxi (Gary) Chen

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EDUCATION

Cornell University	Ithaca, NY
Ph.D. in Computer Science , Programming Languages	Present
Advisor: Prof. Alexandra Silva, Prof. Andrew C. Myers	
University of Pennsylvania, Penn Engineering & The Wharton School	Philadelphia, PA
M.S.E. in Computer and Information Science	2025
B.S.E. in Computer Science , Mathematics minor	2025
B.S. in Economics	2025
Cumulative GPA: 3.97/4.00, <i>Summa Cum Laude</i>	
Advisor: Prof. Steve Zdancewic, Prof. Sanjeev Khanna	

SELECTED PUBLICATIONS

Hanxi Chen, Noam Zilberstein, Andrew C. Myers, Alexandra Silva. Oblivious Probabilistic Outcome Logic: Verifying Probabilistic Programs with an Oblivious Adversary. arXiv, 2026.

Calvin Beck, Hanxi Chen, Steve Zdancewic. Vellvm: Formalizing the Informal. The Eleventh International Workshop on Coq for Programming Languages (CoqPL), 2025.

Calvin Beck, Hanxi Chen, Steve Zdancewic. Vellvm: Formalizing the Informal LLVM (An Experience Report). Nasa Formal Methods: 17th International Symposium (NFM), 2025.

Calvin Beck, Irene Yoon, Hanxi Chen, Yannick Zakowski, Steve Zdancewic. A Two-Phase Infinite/Finite Low-Level Memory Model: Reconciling Integer-Pointer Casts, Finite Space, and undef at the LLVM IR Level of Abstraction. Proceedings of the ACM on Programming Languages, 8 (ICFP), 2024.

RELEVANT RESEARCH EXPERIENCE

Research Assistant , <i>Cornell PL Group</i> , supervised by Prof. Alexandra Silva and Prof. Andrew C. Myers	Sep 2025 –
Cornell University , Computer Science	
- Developing program logics for verifying probabilistic programs under oblivious schedulers, with applications to randomized distributed protocols and hardware. - Building a Lean mechanization library for probabilistic reasoning, including probability-space constructions, outcome triples, and proof automation.	
Research Assistant , <i>PLClub</i> , supervised by Prof. Steve Zdancewic	Jun 2021 – Aug 2025
University of Pennsylvania , Computer and Information Science	
- Contributed to Vellvm in Steve Zdancewic's group, formalizing LLVM IR semantics and low-level memory models in Rocq. - Built Rocq/OCaml differential-testing tools for validating Vellvm's semantics and finding bugs in LLVM implementations.	
Research Assistant , <i>Wharton Research Scholars</i> , supervised by Prof. Sanjeev Khanna	Jan 2023 – Dec 2024
University of Pennsylvania , The Wharton School	
- Studied precedence-constrained knapsack problems, proving pseudo-polynomial algorithms for structured graph classes. - Wrote a 48-page undergraduate technical report and presented the work in the Wharton Research Scholars seminar.	

TECHNICAL SKILLS

Proof Assistants: Lean, Rocq/Coq, Mathlib, QuickChick

Programming: OCaml, Haskell, C/C++, Python, SQL, SystemVerilog

Systems and Tools: LLVM IR, Clang, Dune, Git, Unix, LaTeX

Research Methods: mechanized semantics, program logics, differential testing, property-based testing, randomized algorithms